

# Matt Bowring

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[Website](#) | [Github](#) | [LinkedIn](#)

## Experience

### The MathWorks

Software Engineer (Math/PDE)

MATLAB, C++, Python, CMake, OpenMP, Git

May 2022 - Present

Natick, MA (on-site)

- Lead development of the [MATLAB Quantum Computing Library](#) (unreleased prototype to 9,100+ users)
  - Designed object-oriented MATLAB framework to enable quantum programming across Windows, Linux, and Mac
  - Developed templated C++ state-vector simulator for CPU/GPU with multi-threading using OpenMP
  - Implemented graph-based C++ compiler/transpiler to generate optimized gate-level assembly code
  - Built client-side REST infrastructure to manage async hardware jobs using AWS and IBM cloud services
  - Developed sparse algorithms for estimation of Pauli observables, error mitigation, and gate decomposition
  - Manage CI/CD, write all unit/functional/performance tests, and benchmark against other quantum libraries
- Consult industry users on machine learning and quantum computing applications
  - Implemented quantum neural networks for image classification ([SPIE publication](#)) and reinforcement learning
  - Researched constraint-aware encoding techniques for quantum heuristics using penalty dephasing and quadratization
  - Developed presentations and examples for industry conferences (ACS 25, AWS re: Invent 25, IBM QDC 22-25)

### The MathWorks

Software Engineer

MATLAB, C++, Python, Git

May 2021 - May 2022

Natick, MA (on-site)

- Trained recurrent graph network on QM7-X molecular dataset to predict equilibrium energy and classify optimal configurations; Developed format-agnostic data pipeline to encode atomic properties using OpenBabel
- Developed REST interface to cloud-based quantum annealing hardware; Benchmarked minor-embedding heuristics on constrained combinatorial problems; Wrote parallel simulations to compute qubit transition probabilities for truncated Hamiltonian based on graph-theoretic formalism across coupling strength and energy separation parameters

## Education

### Purdue University (3.8 GPA)

M.S. Mechanical Engineering

Jan. 2025 - Present

West Lafayette, IN (remote)

- Fabricated mixed-signal PCB to solve combinatorial problems using phase dynamics of LC oscillators ([arXiv preprint](#)); Accepted for research mentorship under Stephen Wolfram
  - Developed simulations of coupled oscillators in LTspice and modeled noisy phase dynamics with MATLAB
  - Designed schematics and layouts for two multi-layer boards in Altium Designer and soldered components
  - Developed real-time embedded C++ firmware using SPI and FreeRTOS to measure phases at 1MHz
  - Automated test instrument control to inject perturbations and analyze Fourier spectrum using MATLAB

### The University of New Hampshire (3.9 GPA)

B.S. Mechanical Engineering

Aug. 2017 - May 2021

Durham, NH (on-site)

- Awarded over \$100,000 in scholarships for developing an autonomous quadcopter
- Led the Quadcopter Engineering Team; Served as Makerspace Administrator, Academic Tutor and Mentor

## Projects

### Computer Systems

2024 - Present

- Built custom multi-GPU Linux machine for PyTorch/CUDA research experiments; Configured Beelink home server using NixOS and Tailscale

### Machine Learning

2020 - Present

- Wrote packet parser to process and analyze financial data using Python and Wireshark; Trained recurrent convolutional network to predict dash-camera misalignment using OpenCV and PyTorch; Built data pipeline and Bayesian network to predict trends for online video game using NetworkX and PyTorch

### Robotics

2019 - 2021

- Integrated PX4 flight-stack with Raspberry Pi to enable waypoint tracking for quadcopters; Implemented ROS/MAVROS communication, telemetry, and interfaced Gazebo for SITL simulation with Python; Tuned flight controller and analyzed motor responses using MATLAB; 3D-printed frame, soldered electrical components, and conducted field tests to evaluate controller performance